University of Roehampton London

COURSEWORK 3 ASSESSMENT BRIEF

Programme	MSc Data Science		
Module Title	Machine Learning		
Module Code	CMP-L015-0		
Module Level	7	**Assessment Type(s)**	Report and demo
Word Length / Duration		**% contribution to module mark**	30%
Deadline (date & time) for Submission	13 Feb 2026	**Format/Location of submission**	https://moodle.roehampton.ac.uk/mod/assign/view.php?id=2548510
Assessment Feedback date:	27 Feb 2026		

Learning Outcomes	**Employability and Professional Skills**
This assessment has been designed to provide you with an opportunity to demonstrate your achievement of the learning outcomes listed below. .	This assessment has been designed to provide you with an opportunity to demonstrate your achievement of the following skill(s) which is(are) critical in professional context and jobs. By successfully completing this assessment, you will be able to … 1. Evaluate model performance, compare approaches, and communicate insights clearly through structured reports, tables, and visualisations.

Assessment Requirements

You are required to develop and evaluate a supervised machine learning model using the dataset in ***Assessment*** section of the Moodle page, named as 'cw1data.csv'. Please download the data and import it into your program. The tasks involve end-to-end implementation, from dataset preparation to performance evaluation. Individual steps are:

1. **Import the Dataset**
   - Download the dataset cw1data.csv from the ***Assessment*** section of the Moodle page.
   - Import the dataset into your Python environment.
2. **Data Visualisation**
   - Produce at least one meaningful figure that visualises the dataset.
   - This could be a scatter plot, histogram, correlation heatmap, or any other suitable chart.
3. **Data Preparation**
   - Define the input variables (all columns except y) and the output variable (y).
   - Split the dataset into training and testing sets.
4. **Model Building and Prediction**
   - Implement at least two regression models (e.g., Linear Regression, Decision Tree, Random Forest).
   - Train the models on the training set.
   - Generate predictions on the testing set.
5. **Model Evaluation**
   - Evaluate model performance using suitable regression metrics (e.g., MSE, RMSE, R^2).
   - Present your results clearly in tables and/or figures.
6. **Q & A Section (Compulsory)**
   - Answer questions**:**
     - One or more related to the Python code you developed.
     - One or more related to the concepts discussed in lectures/seminars.
   - **Note:** You will be marked as **NOT PASS** if you fail to answer both questions.
7. **Optional Enhancements**
   - Reproduce your results by rerunning your code during the presentation.
   - Implement at least one additional feature such as:
     - Variable importance analysis
     - Visualising prediction results
     - Performing cross-validation
     - Any other relevant extension

Submission Requirements

1. **Code Submission**

- Submit your Python code file(s) with file name StudentID_MLCW1.ipynb.
- Ensure your design is well-documented with meaningful comments explaining key steps, design choices and result analysis in one page.
- Record your screen while running the codes and include the video clip (file) WITHIN your submission. The video should be less than 10 minutes and highlight the key features you implemented.

You will also need to present your work to a tutor during lab session within two weeks after the due date.

Marking Criteria (90% for artefact; 10% for report)

Grading

Grade	Criteria
100	Excellent work with extra features can be used in the real-world applications with excellent performance in Q&A
92	Excellent work with extra features and minor improvement can be used in the real-world application with excellent performance in Q&A
85	Excellent work with extra features and some improvement can be used in the real-world application with excellent performance in Q&A
82	Excellent work with extra features and improvement can be used in the real-world application with excellent performance in Q&A
78	Excellent work with extra features and major improvement can be used in the real-world application with excellent answers in Q&A
72	Good work completes the required tasks with some additional features that are useful to the real-world applications with good performance in Q&A
68	Good work completes the required tasks with no bugs and code can be used for similar scenarios with excellent performance in Q&A
65	Good work completes the required tasks with no bugs with good performance in Q&A
62	Good work completes the required tasks with minor bugs with good performance in Q&A
58	Good work completes the required tasks with some bugs with sufficient performance in Q&A
55	Complete most required tasks with some adequate performance in Q&A
52	Complete major required tasks with adequate performance in Q&A
48	Complete some required tasks without bugs with poor performance in Q&A
45	Complete half required task with some minor bugs in the complete tasks with poor performance in Q&A
42	Complete half required task with some bugs in the complete tasks with poor performance in Q&A
38	Does not complete less than half the required tasks with poor performance in Q&A

35	Does not complete less than half required tasks and some bugs in the completed tasks with poor performance in Q&A
25	Does not complete less than half required tasks and a lot of bugs in the completed tasks with poor performance in Q&A
15	Submission with 1 task completed with poor performance in Q&A
5	Submission without any tasks completed with poor performance in Q&A
0	No submission

Assessment Success Guidance

To achieve maximum success in this assessment, students should go beyond simply meeting the basic requirements. High marks will be awarded for:

- Clarity and depth of analysis – demonstrate strong critical thinking when comparing models, not just reporting results.
- Professional presentation – ensure your code is clean, well-commented, and your report is concise, well-structured, and visually clear.
- Use of additional techniques – applying optional features such as variable importance analysis, cross-validation, or advanced visualisations can help showcase deeper understanding.
- Independent learning – drawing on online resources, tutorials, and academic references to support your approach.

Academic Misconduct:

"Academic integrity and honesty are fundamental to the academic work you produce at the University of Roehampton. You are expected to complete coursework which is your own and which is referenced appropriately. The university has in place measures to detect academic dishonesty in all its forms. If you are found to be cheating or attempting to gain an unfair advantage over other students in any way, this is considered academic misconduct, and you will be penalised accordingly."

Further details about "Student Code of Conduct" and "Disciplinary Regulations" can be found at:

https://www.roehampton.ac.uk/corporate-information/policies/

Use of Artificial Intelligence (AI)

Please read and apply:

https://roehamptonprod.sharepoint.com/sites/portal/information/LTEU/Documents/4a%20AI%20principles%20and%20guidance%20-for%20Senate.docx

https://roehamptonprod.sharepoint.com/sites/portal/information/LTEU/Documents/4b%20Student%20Guidelines%20on%20the%20use%20of%20AI_.docx

The assessment is designed so that the use of AI during the assessment is possible. You must acknowledge any use of AI and appropriately cite all AI generated outputs. Please make sure you read and understand the assessment guidelines and ask your Module Leader if you have any questions.

**You can find the student guidance on the use of AI in the Library and Nest*

Referencing

References: Referencing any point in your discussion section(s). If you have used ideas from anywhere other than the lecture notes and tutorial examples (e.g., from a book, the Internet, or a fellow student) then include a reference showing where the code or ideas came from and label your code carefully to show which parts are your and which parts are borrowed.

You should include your reference list at the end of your report. You must ensure you use the appropriate IEEE format for your subject area.

Your referencing must use the IEEE referencing style IEEE Citation Guidelines.

It is highly recommended that you use reference management software such as RefWorks that is provided by the university. Your report should have as many references as is required.

It is your responsibility to ensure that you have actually read all the material you reference, and that the references provided in your report are legitimate and NOT AI-generated.

Mitigating circumstances/late penalties

Sometimes circumstances outside of your control may affect your studies and might prevent you from submitting work on time or attending an exam.

The University offers the ability for students to request additional time to complete an assessment or to defer an examination to a later date. If you are finding yourself in such a situation, please speak to your Academic Guidance Tutor, the Roehampton Student Union (RSU) or someone in the **Wellbeing team** first, who can support you. Further details can be found on the mitigating circumstances portal.

If you do not apply for or are not approved for Mitigating Circumstances, late penalties will apply. If work is submitted up to 14 days late, the mark will be capped at 40%; if it is over 14 days late, it will not be marked.

Resubmissions and Reassessment

If you are required to resubmit this assessment or take part in reassessment, you will be notified via Moodle and your student email. Please ensure you check both regularly. Any reassessment tasks will follow the same learning outcomes and criteria.

Submission Checklist

Before you submit, ask yourself:

Have I fully answered the assessment brief?
Have I met the word count and formatting requirements?
Is my referencing complete and accurate?
Have I declared any AI use honestly?
Have I proofread my work?
Am I submitting through the correct platform before the deadline?